DentalAI

An AI-Assisted Dental X-ray Decision-Support System

Design, Implementation, Machine-Learning Model, Clinical Rule Engine and Evaluation

IEEE IC-FTAI Student Project Submission

Technical report prepared for research purposes

Scope statement. This report documents the system as it existed at the time of writing. The first three phases (database schema, scan ingestion with AI detection and dentist review, and the recommendation and contraindication engine) were implemented and tested. The feedback-logging phase (Phase 4) and the React clinician interface (Phase 5) were fully specified but had **not** been implemented; they are reported as designed, not built (Section 11). The clinical rule content was derived from published guidelines and was **not** reviewed by a licensed clinician. DentalAI was designed as an assistive tool and was never intended for unsupervised clinical use.

Abstract

DentalAI was a clinical decision-support system for dental clinics. A dentist uploaded a panoramic radiograph; a three-stage deep-learning pipeline detected pathologies and assigned each to a tooth in FDI notation; the dentist confirmed, rejected or modified every finding; and a deterministic, rule-based engine then proposed treatments and candidate drugs, checked them against the patient's allergies, medications and conditions, and required explicit dentist approval before anything entered the record. The guiding principle was *the system proposes, the dentist decides*.

The backend was built with FastAPI and Supabase (PostgreSQL, row-level security and object storage). Clinical logic was kept in pure, dependency-free Python modules governed by a register of 26 numbered rules, each tied to a published source (ADA 2019, AAPD, AHA 2021, FDA 2013, AGS Beers 2023) or explicitly marked as a project decision. The automated suite comprised 121 tests, and the full workflow was also exercised end to end against a live Supabase instance with synthetic data.

On a held-out set of 23 images (122 detections) the detection pipeline reached a disease-classification accuracy of 97.54%, a tooth-assignment accuracy of 76.23% and a full-match accuracy of 75.41%. This report describes the motivation, architecture, model, data model, clinical engine, verification and limitations.

1. Introduction and Motivation

Panoramic radiographs are routinely taken in dental clinics but are interpreted under time pressure, and subtle lesions such as early caries or periapical radiolucencies are easily missed. At the same time, safe prescribing requires the dentist to hold a patient's allergies, medications, age, weight and comorbidities in mind. DentalAI addressed both problems in one workflow: an AI second opinion on the image, and an auditable safety check on the treatment that follows.

The product was positioned as a business-to-business tool for clinics, which already own the imaging hardware. It was an assistive second-opinion system and not a diagnostic device. Three non-negotiable constraints shaped every design decision:

- **Assistive, never autonomous.** No finding, diagnosis, prescription or dose entered a patient record without explicit dentist confirmation.
- **Deterministic, auditable recommendations.** Dosing and contraindication logic was rule-based and cited; generative models were not permitted to produce drug doses.
- **Clean architectural boundaries.** Data, inference, business logic and presentation were separated so that each could be tested and replaced independently.

2. Development Process

The project was delivered in phases, each ending in a verified, documented state.

Phase	Scope	Outcome
1	Supabase schema, row-level security, storage bucket, seed data	Completed. Six tables, constraints and indexes.
2	Backend steps 1-3: scan ingestion, AI detection, dentist finding review	Completed. Patient, scan and finding APIs; ML client.
3	Backend steps 4-5: recommendation engine, contraindication checker, treatment and prescription endpoints	Completed. 26 rules, 121 tests, live verification.
4	Step 6: feedback logging to *clinical_feedback_logs* and forwarding to the ML service	Specified; not implemented.
5	React + Vite + Tailwind clinician interface	Specified; not implemented.

Between phases, a hardening pass corrected defects found during review. Examples included a reviewer-identifier foreign-key violation caused by a placeholder UUID, a confirmed-findings query that ignored findings the dentist had modified, and a dependency set that failed on a clean install (resolved by pinning *gotrue* below 2.9 and standardising on Python 3.12). A documentation set was maintained so that any new contributor or automated agent could orient quickly: an onboarding guide, an architecture specification, a clinical-guideline register, a database specification and per-phase status reports.

3. System Architecture

3.1 The six-step clinical loop

```
[1] Scan ingestion        panoramic X-ray -> Supabase Storage
[2] AI detection          dentex_week3: quadrants, teeth (FDI), pathology boxes
[3] Dentist validation    confirm / reject / modify each finding
[4] Recommendation        rules map confirmed findings -> treatments + candidate drugs
                          contraindication checker scans allergies, meds, conditions
[5] Dentist confirmation  accept / modify / reject, with gates for warnings and overrides
[6] Feedback logging      audit ledger of AI output vs. dentist decision   (specified)
```

3.2 Components

Layer	Technology	Responsibility
Data	Supabase: PostgreSQL, RLS, Storage (bucket *xray-scans*)	Patients, scans, findings, treatments, prescriptions, feedback logs, image files
Inference	dentex_week3 microservice, port 8001	Detection pipeline; reached only through the backend's *DentexClient*
Backend	FastAPI 0.109, port 8000	Routers, services, pure domain engines
Frontend	React + Vite + Tailwind (planned)	Review UI and decision drawer (not built)

3.3 Backend layering

The backend followed strict one-way dependencies: *api/v1* routers were thin and delegated to *services*, which used pure *domain* models (Pydantic, no I/O). Two engines were deliberately free of Supabase, HTTP and LLM dependencies, so they could be exhaustively unit-tested: *services/contraindication* (catalog, matcher, checker, patient profile) and *services/recommendation* (rules, engine). A single orchestrator, *ClinicalWorkflowService*, composed them with the persistence services using dependency injection, and domain errors (not-found, conflict, rule-violation) were mapped centrally to HTTP 404, 409 and 422.

```
backend/app/
  api/v1/        patients, scans, findings, recommendations, treatments, prescriptions
  core/          configuration, Supabase client factory
  domain/        patient, scan, finding, treatment, prescription, recommendation, errors
  services/
    ml_client/            DentexClient (only path to the ML service)
    supabase_client/      patient, scan, finding, treatment, prescription services
    contraindication/     catalog, checker, matcher, profile, text_match   (pure)
    recommendation/       rules, engine                                     (pure)
    clinical_workflow.py  orchestration + dentist-decision policy
tests/           121 tests, no network required
```

4. Data Model

The schema contained six tables with foreign keys, check constraints and row-level security. The backend used the service-role key to bypass RLS for server operations, leaving the policies in force for any future direct client access.

Table	Purpose
patients	Demographics, weight, allergies, current medications and medical conditions used by the safety checker
scans	Uploaded radiographs, storage path, analysis status and raw model output
findings	One row per AI detection: tooth, disease, confidence, bounding box, and the dentist's review status (pending, confirmed, rejected, modified)
treatments	Proposed or dentist-authored treatments linked to a finding, with decision status and notes
prescriptions	Candidate drugs with dose, frequency and duration, contraindication status, warnings and decision state
clinical_feedback_logs	Audit ledger of discrepancies between AI output and dentist decisions (table created; writer not implemented)

5. The Detection Model

5.1 Pipeline design (dentex_week3)

The deployed model was a three-stage pipeline chosen because a single detector could not reliably tell which tooth a lesion belonged to. The pipeline was served as an autonomous microservice and was never modified by the backend team; the backend consumed only its JSON output.

Stage	Model	Purpose	Settings
1. Coordinator	YOLOv8 Nano	Detect the four dental quadrants	640 px, confidence 0.25
2. Disease detector	YOLOv8	Detect Impacted, Caries, Periapical Lesion and Deep Caries globally	416 px, confidence 0.20
3. Tooth association	Random Forest, 300 trees	Assign each detection to a tooth 1-8 within its quadrant, using 15 spatial and anatomical features	min_samples_leaf 2, balanced class weights, seed 42

The 15 features included quadrant, box centre and size, area, aspect ratio, disease class and anatomically normalised coordinates. Quadrant plus tooth index were converted to two-digit FDI notation (for example tooth 8 in quadrant 4 became FDI 48). The service also generated a plain-text clinical report from the detected facts and exposed endpoints for dentist feedback on findings and on reports, which appended to CSV queues for offline learning.

5.2 Held-out performance

Metric	Value	Basis
Disease accuracy	97.54%	23 held-out images, 122 detections
Tooth-assignment accuracy	76.23%	same
Full-match accuracy (disease and tooth)	75.41%	same

> The held-out set was small (23 images). These figures are indicative and the confidence intervals are wide. Disease accuracy is measured over detections that were found, so it does not by itself describe sensitivity to missed lesions.

5.3 Earlier detector (Phase 1 YOLOv8s) and lessons learned

An earlier five-class YOLOv8s detector was trained on the Tufts Dental Database and the Loki Silvres caries set (80/10/10 split, seed 42; 60 epochs at 1024 px, then 35 fine-tuning epochs without mosaic augmentation). It is retained for reference. Its history produced a concrete methodological lesson: the DENTEX dataset annotated

whole teeth rather than lesions, so a lesion detector trained on it treated real decay as background and Caries recall was 0%. Removing DENTEX and merging lesion-level caries boxes raised Caries mAP@0.5 from 0.009 to 0.648.

Class (final YOLOv8s checkpoint)	Precision	Recall	mAP@0.5
All classes	0.544	0.463	0.380
Caries	0.555	0.667	0.648
Periapical	0.367	0.864	0.553
Non-Odontogenic	0.497	0.450	0.392
Pericoronal	0.302	0.333	0.210
Inter-Radicular	1.000	0.000	0.100 (1 instance)

Documented weaknesses of that checkpoint were extreme sparsity in rare classes, over-prediction of periapical lesions (high recall, low precision, which motivates human review), and loss of the Deep Caries and Impacted classes. The production pipeline above superseded it and restored those classes.

6. Backend Workflow and API

6.1 Scan ingestion and detection (Phase 2)

A scan upload stored the image in Supabase Storage, called the ML service, and converted every detection into a *pending* finding. Findings were never treated as diagnoses until reviewed. The review endpoint accepted confirm, reject or modify; a modification required a valid replacement disease label, and only confirmed or modified findings were passed on to the recommendation step.

6.2 Endpoints

Group	Endpoints
Patients	POST, GET (search, pagination), GET /{id}, PUT /{id}, DELETE /{id}
Scans	POST /upload, GET, GET /{id}, GET /{id}/findings, PUT, DELETE
Findings	GET, GET /{id}, POST /review, GET /scan/{id}/confirmed, PUT /{id}
Recommendations	POST /recommendations/scans/{scan_id} with optional clinical context
Treatments	POST (dentist-authored), GET, GET /{id}, POST /{id}/decision
Prescriptions	POST (dentist-authored), GET, GET /{id}, POST /check (stateless), POST /{id}/decision

7. Recommendation and Contraindication Engine

7.1 Recommendation logic

A rule table mapped each confirmed disease to a treatment plan. Plain caries led to a restorative referral without any antibiotic. Antibiotics were proposed only for diseases where they can be indicated (deep caries, periapical lesion, impacted tooth) and only when the dentist reported active infection through a *ClinicalContext* flag, reflecting ADA 2019 guidance that antibiotics are not indicated for pain alone and never replace source control. Cardiac prophylaxis was proposed for the highest-risk patients undergoing eligible procedures.

Drug selection followed a preference order per indication. For active infection the order was Amoxicillin, then Clindamycin, then Azithromycin; the first candidate that passed the safety check was proposed, and a substitution notice (R-18) was attached whenever the first-line drug had been excluded, so the dentist could see why. Cephalexin was never proposed automatically, because the patient record did not capture the severity of a

penicillin reaction. Duplicate drugs for the same indication on one scan were suppressed and overlapping antibiotics were flagged.

7.2 Dosing

Drug	Adult regimen	Pediatric (under 12)	Source
Amoxicillin	500 mg three times daily, 3-7 days	10 mg/kg per dose proposed (published range 20-40 mg/kg/day, max 500 mg/dose)	ADA 2019; AAPD
Clindamycin	300 mg four times daily; max 1800 mg/day	8 mg/kg per dose proposed (20-30 mg/kg/day, max 450 mg/dose)	ADA 2019; AAPD
Azithromycin	500 mg day 1, then 250 mg daily for 4 days	10 mg/kg day 1, then 5 mg/kg daily	ADA 2019; AAPD
Cephalexin	500 mg four times daily; max 4000 mg/day	10 mg/kg every 6 h	ADA 2019; AAPD (dentist choice only)
Metronidazole	500 mg three times daily, 7 days; add-on after first-line failure	No published regimen; dentist to specify	ADA 2019; AAPD
Ibuprofen	400 mg every 4-6 h; max 3200 mg/day	10 mg/kg capped at 400 mg	ADA; AAPD
Paracetamol	500 mg every 4-6 h; project limit 3000 mg/day, label max 4000	10 mg/kg per dose	AAPD; project limit
Prophylaxis (single dose)	Amoxicillin 2 g; azithromycin 500 mg; clindamycin excluded	Amoxicillin 50 mg/kg (max 2 g); azithromycin 15 mg/kg (max 500 mg)	AHA 2021; AAPD

Pediatric regimens were weight-based, floored to 5 mg for dispensing practicality, capped by single-dose and daily limits, and never allowed to exceed the adult dose. Infection courses carried the ADA follow-up instruction to re-evaluate within three days and to stop 24 hours after symptoms resolve.

7.3 The rule register

Safety checks were implemented as numbered rules with stable identifiers, referenced by tests, API warnings and quality-assurance profiles. Each warning carried a severity (info, warning or contraindicated) and a source citation; the worst severity across rules determined the prescription's status.

ID	Trigger	Severity
R-01 / R-13	Allergy to the drug's class (penicillin to amoxicillin, and the same logic for other catalog drugs)	Contraindicated
R-02	Ibuprofen with an anticoagulant	Contraindicated
R-03	Ibuprofen at age 65 or over (Beers 2023)	Warning
R-04	Single pediatric dose above weight-based or absolute limit	Contraindicated
R-06 / R-15	Paracetamol with hepatic impairment / chronic alcohol use	Contraindicated
R-07 / R-20	Metronidazole with alcohol use / counselling	Warning / Info
R-08	Azithromycin with QT-prolonging drug or QT-risk condition (FDA 2013)	Warning
R-10 / R-11 / R-12	Ibuprofen with active GI ulcer or bleeding / renal impairment / asthma	Contraindicated
R-14	Clindamycin for cardiac prophylaxis (excluded by AHA 2021)	Contraindicated
R-16 / R-17	No published pediatric regimen / pediatric patient without recorded weight	Warning
R-18	First-line drug excluded; substitute proposed	Warning
R-19	Clindamycin: *C. difficile* counselling	Info
R-21 / R-22	Drug not an option for this indication / drug not in catalog	Warning
R-23	Daily dose above published maximum (contraindicated) or project limit (warning)	Contraindicated / Warning

ID	Trigger	Severity
R-24	Metronidazole with warfarin (INR elevation)	Warning
R-25	NSAID with ACE inhibitor or ARB and a diuretic (acute kidney injury risk)	Warning
R-26	Cephalexin with a penicillin allergy on record	Warning

A negative control was also encoded and tested: a sulfa allergy must never trigger a warning for any catalog drug (R-09). Free-text patient data was matched against vocabularies by whole-token sequences, so that, for example, the term "renal" does not match "adrenal".

7.4 Dentist decision gates

- A prescription whose status was *warning* could be accepted only if the dentist set *acknowledge_warnings*.
- A *contraindicated* prescription could be accepted only with a non-blank *override_reason*, which was stored.
- Drugs were re-checked against the current patient record at decision time, so a record edited after proposal could not slip through; substitution notices were preserved.
- Decisions were final; a second decision returned HTTP 409.
- Regeneration was idempotent: untouched proposals were replaced, while treatments and prescriptions the dentist had already acted on were locked and reported as skipped.
- Dentists could also author treatments and prescriptions manually; these were safety-checked in the same way and still required a decision.

8. Verification and Results

Check	Method	Result
Unit and integration tests	pytest with in-memory fakes replacing Supabase; HTTP tests via ASGI transport with dependency override	121 tests passed
Clean-room install	Fresh Python 3.12 virtual environment from requirements.txt with a cleared environment	Passed; byte-compile OK
Catalog self-consistency	Property test: every default regimen passes its own safety check for an adult and a 25 kg child	Passed
Live end-to-end run	Throwaway synthetic patient against the development Supabase: recommendations, regeneration, acknowledge gate, manual creation	Passed; all rows deleted afterwards and verified
Demo branches	Caries, no infection: restorative referral, 0 drugs. Periapical lesion with infection: amoxicillin 500 mg q8h. Same with penicillin allergy: R-18 warning and clindamycin	Passed

A transient read timeout occurred once during live testing and did not recur on retry. No real patient data was used at any point.

9. Clinical Evidence Base

All clinical values were taken from published guidance and recorded in a guideline document with a per-rule evidence status. The ADA 2019 antibiotic guideline (chairside guide) and the AAPD *Useful Medications* tables were read in full. The AHA 2021 prophylaxis update, the FDA 2013 azithromycin communication, the AGS Beers Criteria 2023 and the metronidazole-warfarin and NSAID "triple whammy" literature were confirmed from secondary summaries because full text was not accessible. Decisions made conservatively by the team were labelled **PROJECT** (for instance treating any asthma as an ibuprofen contraindication, and the 3000 mg/day paracetamol limit). The contribution policy required every future rule change to cite a published source or carry that label.

10. Limitations

- **No clinician review.** The rule content was source-backed but never validated by a licensed dentist or pharmacist.
- **Not modelled:** pregnancy and lactation, renal dose adjustment, severity of liver or kidney disease, and the severity of past penicillin reactions, because the schema had no fields for them.
- **No authentication.** Reviewer and dentist identifiers were left null; the system ran with a single service credential.
- **Small ML validation set.** Twenty-three images and 122 detections limit the strength of the reported accuracy.
- **Single-source geography and equipment.** No evidence was gathered on performance across different X-ray machines or clinics.
- **Catalog scope.** Seven drugs were covered; others produced an explicit "not checked" warning (R-22).
- **Sources.** Several secondary sources were confirmed from summaries only.

11. Designed but Unbuilt Components and Future Work

Phase 4, feedback logging. The design wrote one row to *clinical_feedback_logs* for every finding review, treatment modification and prescription override, recording the original AI value, the dentist's value and the reason, and forwarded corrections to the ML service's feedback endpoints for the offline retraining queue. The table and the ML endpoints existed; the writer did not.

Phase 5, clinician interface. The design specified a React, Vite and Tailwind application with a scan viewer overlaying bounding boxes, a finding review panel, a treatment and prescription decision drawer, and a persistent "Pending Dentist Validation" badge on every AI-generated item. A requirement was that no clinical logic be duplicated on the frontend.

Further recommended work: clinician validation of the rule register; authentication and per-dentist audit trails; extending the patient record with pregnancy and renal function; a larger multi-clinic validation set with sensitivity analysis; and prospective study of whether the tool changes dentist behaviour.

12. Conclusion

DentalAI demonstrated that an assistive dental workflow can combine an image-based detector with a transparent, rule-based prescribing safety layer while keeping the dentist in control at every step. The backend through Phase 3 was implemented, tested and exercised against a live database; the detection pipeline achieved 97.54% disease accuracy and 75.41% full-match accuracy on a small held-out set. Clinical validation, feedback logging and the user interface remained to be completed before any real-world use.

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• Tufts Dental Database; Loki Silvres caries dataset; DENTEX Challenge 2023 (dataset granularity discussed in Section 5.3).